

I: THE GATEWAY TO THE DIGITAL FRONT-LINE

1.1 Introduction

In an era where data is more valuable than gold, protecting digital assets has become a global priority. The Cyber Security program at Duy Tan University (DTU) is designed to transform students into elite defenders of the cyberspace. This program isn't just about theory; it is a battle-tested curriculum aimed at neutralizing sophisticated cyber threats.

1.2 Mission and Vision

DTU aims to provide a world-class education by collaborating with global leaders like Carnegie Mellon University (CMU). Our vision is to produce graduates who possess:

- **Technical Excellence:** Mastery of hacking tools and defense mechanisms.
- **Ethical Integrity:** A commitment to using skills for the benefit of society.
- **Global Readiness:** Capabilities that meet international industry standards.



II: CAREER FRONTIERS & PROFESSIONAL EVOLUTION

2.1 The Modern Security Workforce

The "Cyber Gap" is real—millions of jobs remain unfilled worldwide. Graduates from DTU are prepared to step into high-stakes roles immediately:

- SOC Analyst (Security Operations Center): The "First Responders" who monitor networks 24/7 for suspicious activity.
- Penetration Tester (Ethical Hacker): Professionals hired to legally "break into" systems to find weaknesses before criminals do.
- Digital Forensic Examiner: The "Cyber Detectives" who investigate data breaches and recover evidence for legal cases.

2.2 Career Growth Roadmap

A career in security is a lifelong journey of learning:

1. Junior Level: Focused on technical implementation and monitoring.
2. Mid-Level: Designing secure architectures and leading incident responses.
3. Executive Level (CISO): Managing the entire security strategy and budget for a global corporation.

III: THE ACADEMIC BLUEPRINT: WHAT YOU WILL MASTER

3.1 Integrated Curriculum

The curriculum at DTU is divided into four critical pillars to ensure a 360-degree understanding of the field:

- Foundation: Networking, C/C++ Programming, and Discrete Mathematics.
- Systems: Linux/Windows Administration and Database Security.
- Offensive: Ethical Hacking, Exploitation Techniques, and Social Engineering.
- Defensive: Encryption, Firewall Management, and Incident Response.

3.2 Hands-on Learning (The 70/30 Rule)

At DTU, we believe in Learning by Doing. 70% of the time is spent in the lab, simulating real-world attacks in a controlled environment (Cyber Range).

Year	Focus Area	Key Mapped Course
Year 1	The Basics	Intro to IT & Programming
Year 2	Infrastructure	Network Security & OS Hardening
Year 3	Attack/Defense	Ethical Hacking & Cryptography
Year 4	Specialization	Digital Forensics & Capstone Project

 70% HANDS-ON LAB & PRACTICE		 30% THEORY & CLASSROOM
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IV: THE CMU ADVANTAGE & GLOBAL RECOGNITION

4.1 The CMU Partnership

DTU is one of the few universities in Vietnam to adopt the curriculum of Carnegie Mellon University (USA), the world leader in Cyber Security. This means students study the exact same materials used by top-tier students in the United States.

4.2 Industry Certifications

The program is mapped to prepare students for "Golden" industry certifications during their studies:

- CompTIA Security+: Validating foundational security knowledge.
- CEH (Certified Ethical Hacker): The industry standard for offensive security.
- CISSP: Preparation for high-level management roles.

4.3 Conclusion

Choosing An Ninh Mạng at Duy Tan University is not just choosing a major; it is choosing to become a guardian of the digital world. With our advanced facilities and international curriculum, your future is secure.